(1)

(Please write your Enrollment Number) Enrollment No. 04801172022

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OFFLINE)
Course Name: B.Tech (CSE-AI /ECE-AI), Semester: VIII
(May, 2026)

Subject Code: BAI 416	Subject: AI in Healthcare
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)
a)	Define "Structured" and "Unstructured" data	(2.5)
b)	State two historical milestones in the evolution of AI within the healthcare sector.	(2.5)
c)	What is a Clinical Decision Support System (CDSS)?	(2.5)
d)	Differentiate between image "classification" and "segmentation" in medical imaging	(2.5)
e)	How can Natural Language Processing (NLP) improve Electronic Health Record (EHR) analysis	(2.5)
f)	Name two common predictive modeling techniques used for patient risk assessment.	(2.5)
g)	What is the primary purpose of HIPAA compliance in AI healthcare applications?	(2.5)
h)	Mention one major ethical dilemma associated with using AI for medical diagnosis.	(2.5)
	UNIT I	
Q2	Describe the typical workflow of a healthcare data analysis project, from data collection to deployment."	(10)
Q3	Why is data preprocessing and cleaning considered the most critical step before performing hands-on analysis with healthcare datasets?	(10)
	UNIT II	
Q4	Describe the technical process of medical image segmentation and classification, and explain how these techniques assist in early disease detection?	(10)
Q5	How AI-powered diagnostics have transformed traditional CDSS, and analyze the ethical considerations involved in their implementation?	(10)
	UNIT III	
Q6	Describe the end-to-end process of developing a predictive model for patient risk assessment	(10)
Q7	Explain the significance of Natural Language Processing (NLP) in extracting value from Electronic Health Records (EHRs). Discuss the specific NLP tasks involved and the ethical considerations unique to processing medical text.	(10)
	UNIT IV	
Q8	How can healthcare organizations address the issue of patient data protection'?	(10)
Q9	What are the potential benefits and the technical hurdles that must be overcome for widespread adoption AI-driven Telemedicine and Genomics?	(10)